ISLAMIC UNIVERSITY OF TECHNOLOGY
A Subsidiary Organ of The Organisation of Islamic Cooperation (OIC)

Final Project Report

Airport Management System

Supervisor : Faisal Hussain

Team Members

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SPL-I

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Problem Statement

Airport operations involve multiple interdependent activities such as flight scheduling, gate and runway allocation, passenger booking, check-in, and boarding. When these activities are handled manually or through disconnected processes, they often result in scheduling conflicts, inefficient use of resources, delays in operational decisions, and poor passenger service during disruptions. In addition, weather-related changes and limited infrastructure make airport coordination more difficult. Therefore, there is a need for a centralized system that can manage airport workflows in an organized, secure, and efficient manner.

Potential Solution

The proposed solution is a simulation-driven Airport Management System that integrates the core activities of a small airport into a single terminal-based application. The system is designed to automate flight scheduling, passenger booking and check-in, gate and runway allocation, role-based access control, and weather-aware operational decisions. By bringing these modules together in one platform, the system reduces manual dependency, improves coordination among airport functions, and creates a more reliable operational workflow.

Implemented Features

The developed system includes several core features that support airport operations. It provides **flight management**, where flight information, status, passenger count, and operational conditions are maintained systematically. It also includes **resource management** for the allocation of gates and runways so that infrastructure conflicts can be minimized. A **booking and passenger service module** allows flight selection, seat handling, booking confirmation, check-in, and

boarding pass generation. For security, the project implements **role-based authentication**, along with password validation and hashed password handling. The system further supports **weather simulation and weather-aware rescheduling**, allowing operational decisions to adapt to changing conditions. Finally, the project uses **file-based persistence and reporting**, which stores flight, passenger, and login data across sessions and supports operational reporting in text-based format.

Tools and Technology

Programming Language: Java
(The core programming language used to develop the system, implementing object-oriented design for better modularity and maintainability.)

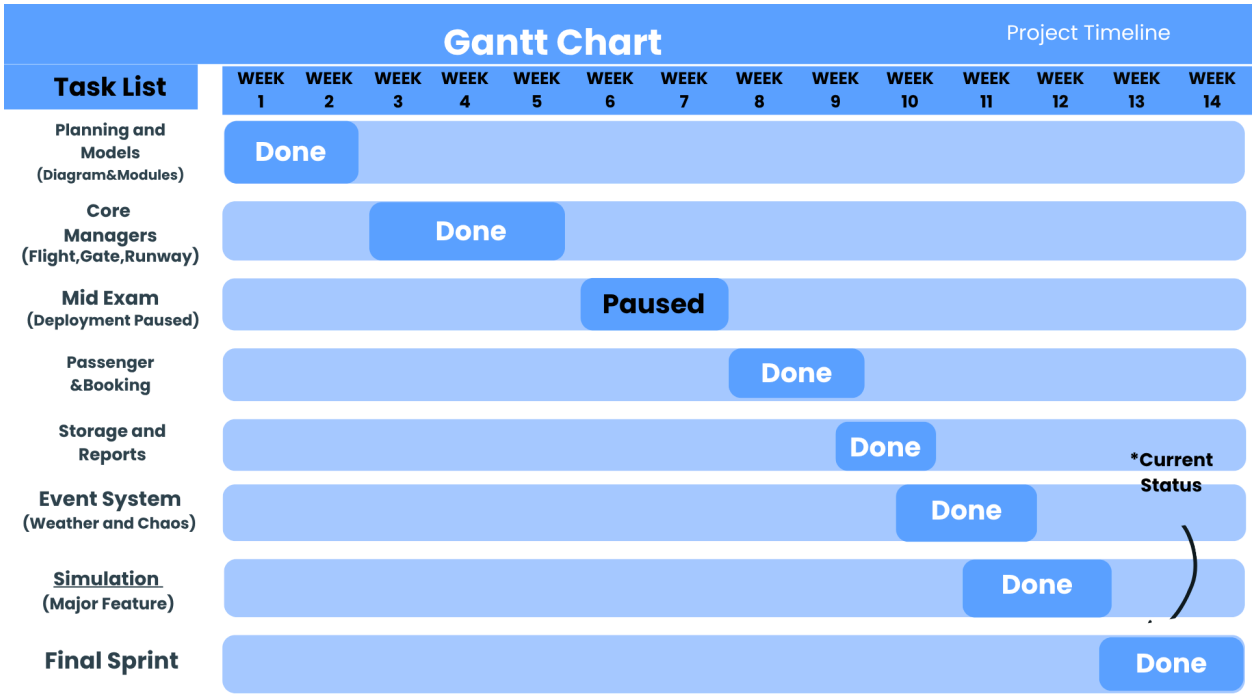
Development Environment: IntelliJ IDEA
(The Integrated Development Environment (IDE) used for coding, debugging, and ensuring clean architecture in the project.)

Version Control: GitHub
(A platform for managing code versions, facilitating collaboration, and tracking changes across the development process.)

Application Domain

This project belongs to the domain of **airport operations management** and **transportation system software**. More specifically, it is suitable for small-airport simulation, academic demonstration of resource allocation and workflow automation, and training-oriented environments where flight operations, passenger handling, and infrastructure coordination must be represented in a simplified but functional manner.

Project Timeline



Contribution

230042115-Sheikh Alimun Alahi's Contributions

- **Gate and Runway Management System**
  (Developed the system responsible for managing the allocation and status of gates and runways.)
- **Seat Layout**
  (Designed the seating arrangement for flights, ensuring proper seat assignments and capacity handling.)
- **Console Layout**
  (Created the user interface layout for the command-line terminal to improve user experience.)

- **Weather Type and Weather Slot Features**
  (Implemented weather type selection and slot management, which influence flight scheduling and operational decisions.)
- **Simulation Engine**
  (Developed the engine driving the simulation, including time updates, weather changes, and event handling.)

230042128-Rakibur Rahman Ananta's Contributions

- **Passenger Management**
  (Developed the system for storing and managing passenger data, ensuring effective check-ins and boarding.)
- **App User and Authentication**
  (Implemented user management and authentication systems to ensure secure access to various modules.)
- **Password Security Utilities**
  (Created password encryption and validation mechanisms to ensure secure user login and data integrity.)

230042147-Irfan Sadik Chy's Contributions

- **Flight and Flight Management**
  (Designed and managed the system for flight scheduling, status tracking, and operational conditions.)
- **Departure and Arrival Handling**
  (Implemented the systems that handle flight departures and arrivals, ensuring accurate scheduling and monitoring.)
- **Booking Management**
  (Developed the booking system that handles seat selection, ticket generation, and booking confirmations.)
- **Weather Manager**
  (Created the module for managing real-time weather conditions that impact flight clearance and scheduling.)

Future Work :

The project can be extended in several important directions. First, it can be integrated with **real-time airline APIs** to support live operational data. Second, a **mobile application for passengers** can be developed to improve accessibility and user interaction. Third, an **AI-based delay prediction system** can be introduced to improve forecasting and operational planning. In addition, since the current system uses file-based storage, future versions may be enhanced with a **database-backed architecture** to improve scalability, data consistency, and real-world applicability.